

B

CSE422, Lab MID

Time - 60 Minutes

Name: _____ ID: _____ Section: _____

Question 1: 6 Marks

A rescue robot needs to navigate through a disaster zone to reach a survivor. The zone is represented as a 5x5 grid where:

- 'R' is the robot's starting position at (0, 0)
- 'S' is the survivor's position at (4, 4)
- '0' represents paths

Grid:

```
R 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 0
0 0 0 0 S
```

Your task is to implement A* search to find the shortest path cost from R to S, considering the following:

- Movement: Up, Down, Left, Right only
- Cost: Each move costs 1
- Heuristic: Manhattan distance

Input: Start (0,0), Goal (4,4)

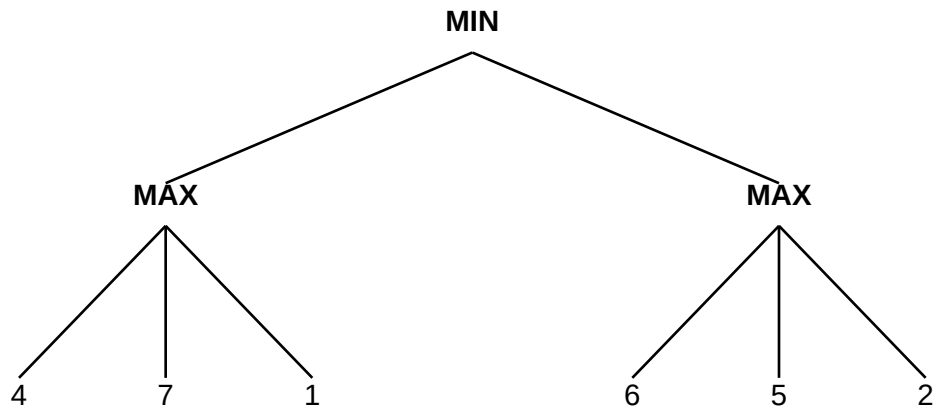
- You do not need to take user input.
- You can represent the graph using any data structure. You can simply use the matrix as well.

Question 2: 3 marks

A simple turn-based game produces a list of possible final scores. Player A (MIN) wants to minimize the final score. Player B (MAX) wants to maximize the score.

The game score list is: [4, 7, 1, 6, 5, 2]

The game tree looks like:



Your task is to determine the final score Player A can guarantee.

- You do not need to create the tree. You can work with the score list given in the question.

Once you are done, submit your code using the submission form link.